which is formed by taking the differences in yields between liquid newly issued government bonds (on-the-run) and relatively illiquid seasoned government bonds (off-the-run). Higher values indicate more illiquid markets. Illiquidity spreads widened in 1998 as a result of the Russian default crisis and the failure of Long-Term Capital Management, a large hedge fund (see chapter 17). The second dramatic widening at the end of the sample is due to the financial crisis. The illiquidity spread grew dramatically in the latter part of 2008 after the failure of Lehman Brothers and reached its greatest extent in January 2009. As the financial system stabilized over the last few months in 2009, illiquidity risk fell. The volatility risk factor in Panel C represents returns from selling volatility (see chapters 1 and 7). Returns from the volatility strategy fell off a cliff in 2008, wiping out ten years of cumulated gains. The negative active returns during the 2008–2009 financial crisis coincided with both more illiquid markets (Panel B) and pronounced losses on short volatility strategies (Panel C).

In Panel D of Figure 14.3, I ask whether the negative active returns over 2008–2009 could have been anticipated. This is a different question than whether the negative active returns could have been perfectly forecast. ("Prediction is very difficult, especially if it's about the future," said Nobel laureate physicist Nils Bohr.) The figure graphs the actual active returns each month in dots and the active returns explained by factor exposures in the solid lines. There are two lines: one estimates the factor exposures over the full sample and the other uses rolling windows. The latter updates the factor exposures each month and uses information available only at the prevailing time. Remarkably, the two lines almost coincide.[12] The two lines account for much of the variation in actual active returns: they track the fairly stable active returns prior to the onset of the financial crisis, they follow down the large losses during the financial crisis, and factors also explain the bounce back in active returns in late 2009 when markets stabilize.

My interpretation of Panel D is that, had the factor exposure of the fund been properly communicated prior to 2008, the active losses during the financial crisis would not have caused such a strong reaction in the Norwegian public. Had the responsible parties warned that sooner or later markets always become illiquid and volatility shoots upward (but that bearing such risks is worth it, and things eventually turn out fine), then the negative active returns in 2008 might have been within anticipated loss limits. The Norwegians did not find the losses from equity positions, however large, surprising because they were aware of the downside risk.

The Professors' Report showed that Norway's active returns have large exposure to systematic, dynamic factors that are appropriate for Norway because the factors earn risk premiums over the long run. The factor exposures, however, did not come about through a process that deliberately chose which factor premiums should be harvested and optimally determined the sizes of the factor

[12] Interestingly, the estimation using the expanding window predicts much more severe losses in September 2008 than what actually occurred. This is consistent with NBIM's skill in managing time-varying factor exposures by changing either internal or external investment strategies.